

Technical Assessment Task

Title

Design an AI-Driven OSINT Threat Monitoring and Risk Assessment Prototype

Background

The KTP project aims to develop an AI-based intelligence system to automate open-source intelligence (OSINT) collection, analysis, and risk assessment to support proactive threat monitoring.

As part of the KTP Associate role, the successful candidate will be expected to design scalable and intelligent solutions that combine:

- AI / machine learning
- Cyber security knowledge
- Structured and unstructured data analysis

Task overview

You are asked to design a prototype solution for an AI-driven OSINT threat monitoring system.

Your solution should demonstrate how open-source data can be:

- Collected
- Processed
- Analysed
- Used to generate actionable threat intelligence

The project does not need to be production-ready. The focus is on your technical reasoning, problem-solving approach, and ability to communicate your design decisions clearly.

Suggested public datasets / data sources

Candidates may use any suitable public datasets or open-source data sources. Recommended sources include:

News & OSINT sources

- NewsAPI: <https://newsapi.org/>
- Google News RSS: <https://news.google.com/rss>

- **The Guardian Open Platform**: <https://open-platform.theguardian.com/>

Cyber security & threat intelligence sources

- CISA Cybersecurity Advisories: <https://www.cisa.gov/news-events/cybersecurity-advisories>
- MITRE ATT&CK Framework: <https://attack.mitre.org/>
- CVE Vulnerability Database: <https://www.cve.org/>

Candidates may also use other publicly available sources where appropriate.

Specific project requirements

1. System design

Propose an architecture for the system (diagram encouraged), including:

- Data sources
- Data ingestion methods
- Storage and processing pipeline

2. AI / Analytical approach

Describe methods used to:

- Detect relevant threats
- Extract key entities or signals
- Assess threat severity or risk levels

3. Prototype / Demonstration (Lightweight)

Provide a small-scale implementation or pseudo-code, workflow, or sample outputs. This is flexible and subject to your understanding and capability.

Examples may include:

- A simple script
- Pipeline design
- Sample outputs
- Notebook or prototype

4. Risk assessment framework

Propose a method to:

- Prioritise threats

- Assess risk levels
- Generate a simple “threat report” output

5. Discussion of key considerations

Briefly reflect on:

- Data quality and limitations
- Ethical and legal considerations
- Scalability and deployment challenges
- Potential future improvements

Project deliverables

1. Written report

Please prepare a report of a maximum of 10 pages (Please do not exceed this limit) covering:

- System design and architecture
- Methodology and technical approach
- AI / analytical methods used
- Risk assessment framework
- Key assumptions and limitations
- Potential future improvements

2. Presentation

During the interview, please prepare a short presentation (max 10 minutes) summarising:

- Your understanding of the problem
- Proposed solution and architecture
- Key technical decisions
- Main findings, limitations, and future improvements